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Feeding preweaning Holstein calves a synbiotic supplement increases their energy-corrected milk yield as lactating cows

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ABSTRACT

Prebiotics and probiotics are feed additives that can benefit the host by modulating the gut microbiome, which is crucial in digestion, immunity, and overall animal health. This study aims to evaluate the effects of supplementing prebiotics, probiotics, or synbiotics to preweaning Holstein calves on their future milk yield. This study is a retrospective analysis of milk yield records from dairy cows that were randomized at birth to 1 of 4 twice-daily treatments administered during the preweaning period: (1) control, no additive (CON), (2) prebiotic (PRE; 7 mL of *Saccharomyces cerevisiae* yeast culture), (3) probiotic (PRO; *Bacillus subtilis* and *Lactobacillus plantarum*, delivering ~1 billion and 250 million cfu per head per day, respectively), or (4) synbiotic (SYN; combination of both PRE and PRO at the same dosages as the PRE and PRO treatments). The study was conducted on a dairy farm in Fresno County, California, between 2019 and 2023, involving 1,296 Holstein cows over their first 3 lactations for a total of 2,735 lactations. Monthly test day records for milk yield, fat, and protein were used to calculate ECM, standardized to 4% fat and 3.3% protein, totaling 26,464 monthly test day milk records. A 2-piece splines mixed-effect regression model evaluated the effect of treatments on ECM yield. For the first lactation, ECM yield was estimated at 28.66 kg on the first DIM, peaked at 42.1 kg, and declined to 21.34 kg by 305 DIM. For parity ≥2, ECM yield was 41.06 kg at 1 DIM, peaked at 54.2 kg, and 33.74 kg at 305 DIM. The SYN treatment increased ECM yield by 1.00 kg/d compared with CON. This increase was primarily due to an increase in milk fat yield, with 0.048 kg/d more fat produced compared with the control group. No differences in ECM yield between PRE, PRO, or CON were observed. These findings suggest that supplementing SYN during the preweaning period increased milk, milk fat, and ECM yield across lactations 1, 2, and 3.
Key words: preweaning Holstein dairy calves, synbiotic, milk yield

INTRODUCTION

Dairy cows host a complex ecosystem of microorganisms within their digestive system. These microorganisms include bacteria, protozoa, fungi, and viruses that play vital roles in digestion, nutrient absorption, metabolic regulation, and immune function (Liu et al., 2021; Xu et al., 2021). A balance between the microorganisms in the gut is essential for optimal health, growth, and production. Any disruption of this balance can lead to gastrointestinal disorders, weak immunity, and increased vulnerability to diseases (Sanz-Fernandez et al., 2020; Welch et al., 2022) and may have lasting consequences for future milk yield. For example, diarrhea during the preweaning period reduced future milk yield in a study on 2,272 female calves. Calves that had a history of diarrhea during the preweaning period produced 325 kg less 305-d mature-equivalent milk in the first parity compared with unaffected calves (Abuelo et al., 2021).

Prebiotic and probiotic feed additives can enhance cows' health and productivity, and they can play a role in preventing disease, leading to improved growth and increased milk production. (USDA, 2012; Leistikow et al., 2022; Várhidi et al., 2022). Optimum growth for dairy calves during the preweaning period plays a vital role in determining their future milk yield. Greater ADG in preweaning calves is associated with increased future milk yield (Zanton and Heinrichs, 2005; Gelsinger et al., 2016). In Chester-Jones et al. (2017), calves with higher preweaning ADG produced more milk fat and protein in their 305-d first parity compared with the calves with lower ADG. For every 1-kg increase in ADG, milk fat and protein yields increased by 27.3 kg and 26.1 kg per adjusted 305-d lactation, respectively.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Table 1. Cutoff points for potential outliers in unadjusted milk yield by lactation for all treatments

Lactation	Milk yield (kg/d)	SD[1]	SEM	IQR[2] (kg/d)	50th percentile (kg/d)	75th percentile (kg/d)	Skewness
1	35.2	9.9	0.01	12.6	36.7	42.3	−0.59
2	39.8	25	0.2	22.1	41.7	50.9	−0.26
3	44.6	14	0.04	20.1	46.1	55.1	−0.33
4	50.2	13	0.1	17.0	52.2	59.7	−0.61
Multiparous	42.0	14	0.02	21.5	43.7	52.8	−0.29

[1]Standard deviation of daily milk yield values from the mean.

[2]IQR = interquartile range. The IQR and 75th percentile values were used to calculate the cutoff for milk yield outlier records in the first parity. The mean of IQR and 75th percentile for parity ≥2 was used to calculate the cutoff for milk yield outliers in multiparous records.

Prebiotics and probiotics such as *Saccharomyces cerevisiae* yeast culture, *Bacillus subtilis*, and *Lactobacillus plantarum* enhance the growth of beneficial bacteria in the gut (Gibson and Roberfroid,1995) and can improve health and growth of preweaning calves. In a study by Lucey et al. (2021), calves supplemented with prebiotics (*Saccharomyces cerevisiae* enriched with mannan-oligosaccharide) and synbiotics (*Saccharomyces cerevisiae* enriched with mannan-oligosaccharide and *Bacillus subtilis*) had increased ADG from 42 to 56 d compared with control (85 g and 78 g, respectively). Probiotic calves had 100 times lower fecal shedding of *Cryptosporidium* oocysts at 14 d, and calves fed a prebiotic had a lower quantity of *Escherichia coli* and pathogenic *E. coli* fecal shedding compared with the control calves.

Early-life nutrition and health influence future milk yield, but little is known about the long-term influence of nutritional additives such as prebiotics and probiotics. Because most studies focus primarily on the short-term effects of prebiotic and probiotic additives during the preweaning period, there is a gap in our understanding of the potential long-term benefits of these additives on milk production. In this study, we will follow calves from the Lucey study (Lucey et al., 2021) to assess the long-term effects of preweaning supplementation with pre-, pro-, or synbiotic on future milk yield. This will be accomplished by analyzing daily milk, protein, and fat yields and monthly ECM yield from these calves as lactating adult cows.

We hypothesize that supplementing Holstein calves with prebiotic (*Saccharomyces cerevisiae* yeast culture), probiotic (*Bacillus subtilis* and *Lactobacillus plantarum*), or a combination of both (synbiotic) during the preweaning period will increase calves' future milk production and health.

MATERIALS AND METHODS

All procedures for the original calf study were approved by the University of California Davis Animal Care and Use Committee (protocol number 20291). No approval from the University of California Davis Animal Care and Use Committee was needed for this study, as it was conducted using herd records from cows that were enrolled in the Lucey et al (2021) study as calves.

Study Design and Population

A retrospective cohort study was conducted based on data from a randomized clinical preweaning calf study between January 5 and July 2, 2018, on a dairy farm in Fresno County, California. The original study included 1,801 preweaning Holstein calves randomly assigned to 4 treatments: 450 control calves (**CON**; no supplements), 450 calves fed prebiotic (**PRE**; 7 mL of *Saccharomyces cerevisiae* yeast culture), 451 calves fed probiotic (**PRO**; *Bacillus subtilis* and *Lactobacillus plantarum*, delivering ~1 billion and 250 million cfu per head per day, respectively), and 450 calves fed synbiotic (**SYN**; a combination of both PRE and PRO at the same dosages as the PRE and PRO treatments).

Cows were raised in freestall barns and milked twice daily in 2 rotary parlors with daily milk recording. Data were collected from January 2019 to June 2023, using 34 backups from the herd management software DHI-Plus (Amelicor, Provo, UT). The herd was composed of 7,780 Holstein cows, including 7,006 lactating and 774 dry cows when the lactation records were collected (i.e., calves were lactating adults).

Individual cow milk fat and protein yields were measured monthly by the Fresno County DHIA and used in modeling the effect of treatment on milk yield.

After weaning, there were no differences in herd management between treatment groups of heifers; all heifers received the same TMR formulations and were exposed to the same management conditions. Cows were moved between pens by farm staff based on DIM, milk yield, and reproductive status. Two TMR (high- and low-milking) were fed to the lactating cows. All pens had consistent stocking densities, cooling systems, and milking and feeding times.

Table 2. Assessment of bias due to culling and death events in the adult cows (n = 1,296) by treatment and last lactation

Factor	Coefficient	SE	*P*-value	95% CI
Treatment				
Control	Referent			
Prebiotic	0.15	0.2	0.5	−0.30, 0.59
Probiotic	0.18	0.2	0.4	−0.26, 0.62
Synbiotic	−0.10	0.2	0.7	−0.52, 0.32
Final lactation				
1	Referent			
2	−1.11	0.2	<0.05	−1.5, −0.75
3	−2.15	0.2	<0.05	−2.6, −1.7
Constant	2.29	0.2	<0.05	−1.9, 2.6

Data Collection

Data were extracted from the herd management software program DHI-Plus and then exported into an Access database (Microsoft 365 suite by Microsoft Corp., Redmond, WA). Statistical analysis was conducted using Stata BE 18 (StataCorp LLC, College Station, TX).

Inclusion and Exclusion Criteria

Milk records included lactating cows that had completed the original calf trial with a consistent lactation record timeline; that is, daily milk records corresponding to the days in milk, the first milk record starting on the first day after calving, and the last milk record on the day of dry-off, without duplicate entries. Inconsistent records were excluded. Records from cows that had disease but were not culled or died were included in the final datasets. Cow records that contained outliers were excluded.

Missing milk records were quantified by the total number of milk records subtracted from the total DIM for each cow and parity. Each cow had 1 milk record per day from calving to dry-off. The DIM was calculated as the difference between calving and dry-off date. Data for the 2-wk period from July 31 to August 14, 2021, were lost due to a computer and milking equipment error. Cows with missing records were not excluded from the datasets.

Statistical Analysis

All statistical analyses were conducted in Stata BE 18.0 (StataCorp LLC, College Station, TX). A 5% level of significance was observed. The study unit was the individual cow.

There are 2 sets of estimates of milk production records: daily milk production generated from the milking equipment and monthly milk production records from monthly DHIA tests. We extracted both from the dairy herd management software backup files. The adjusted datasets are estimates of milk production on monthly milk records from DHIA milk test days, which include milk components, namely percent fat, protein, and solids nonfat, that can be used to adjust milk for fat and energy contents. The nonadjusted dataset comprises daily milk records from the milking equipment. Both datasets exclude outliers and inconsistent records. For the adjusted dataset, milk protein, fat, and milk yields were converted to ECM to standardize milk production. The ECM was calculated using the actual monthly milk fat (%) and protein (%) values obtained from the Dairy Herd Improvement Association monthly records. Each cow's corresponding milk, fat, and protein values were used to calculate ECM. The ECM were standardized to a reference milk composition of 4% fat and 3.3% protein using the following formula (Hall, 2023):

$$\text{ECM} = [\text{kg Milk} \times (0.383 \times \text{Milk fat \%} + 0.242 \times \text{Milk crude protein \%} + 0.7832)/3.1138].$$

Outlier Detection in Milk Yield Data

Tukey's method was used to calculate the upper bound of milk yield for each parity to identify outliers (Table 1). Milk yields above the upper bound were flagged, as they were deemed above the expected range of possible milk yield. The lower bound for milk yield was not calculated, as factors such as illness and estrus (heat) in cows could cause a decline.

The upper bound for identifying outliers was calculated as follows:

$$\text{Upper bound for the first parity} = \text{Q3} + (1.5 \times \text{IQR}) = 42.3 + (1.5 \times 12.6) = 42.3 + 18.9 = 61.2 \text{ kg},$$

where **Q3** is quartile 3 or 75th percentile (42.3) and **IQR** is interquartile range (12.6). Using Tukey's method, the upper bound was calculated as follows:

$$\text{Upper bound for second or greater parities} = \text{Q3} + (1.5 \times \text{IQR}) = 52.8 + (1.5 \times 21.5) = 52.8 + 32.25 = 85.1 \text{ kg}.$$

The normality of the milk variables (ECM milk, monthly milk fat yield, monthly milk protein yield, and nonadjusted daily milk yield) was checked after the removal of outliers using quantile and normal probability plots before building the mixed-effect regression model. Models were built using manual forward selection, and best-fit models were chosen using the lowest Akaike information criterion (**AIC**) and likelihood ratio test (**LRT**) <0.5.

Model residuals were analyzed using histograms and quantile–quantile plots to evaluate normality. Addition-

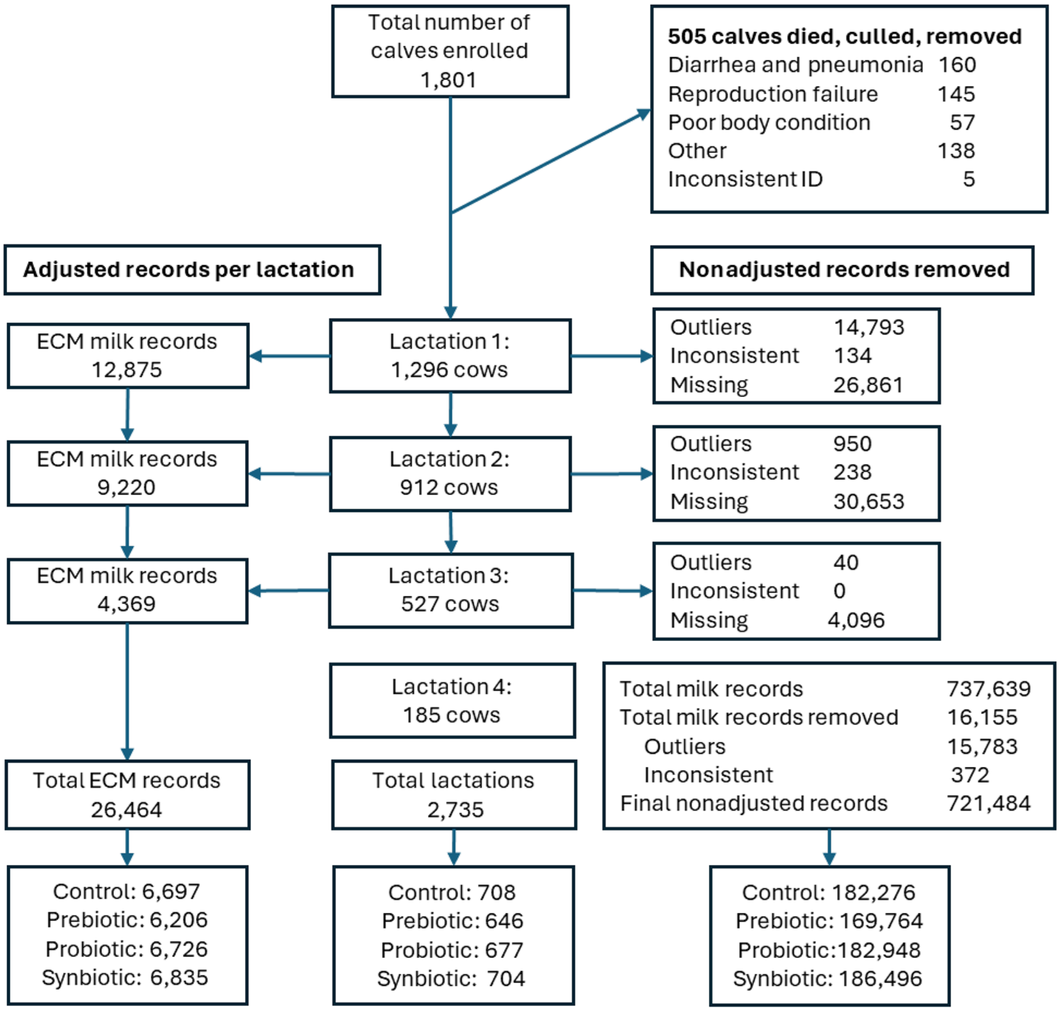

Figure 1. Flowchart illustrating the progress of 1,801 calves from calving through parities, illustrating the exclusion reasons in each step and ending with the final number of records for final analysis, distributed across 4 treatments.

ally, comparisons between homogeneous and heterogeneous residual variances were compared.

Assessing Loss of Follow-Up Bias

A logistic regression model was specified to investigate the association between treatment group (independent variable) and culling or death (dependent variable) between first calving and removal from herd or censoring (n = 1,296 cows).

Mixed-Effect Regression Models

Treatment effects were contrasted using 2 parameterizations: (1) as the conventional mutually exclusive treatment groups PRE, PRO, and SYN contrasted to the referent group CON in a 2-piece spline model for effect estimation of each of the treatments including SYN, as a distinct biological intervention, making no assumption that its effect must equal the sum of PRE and PRO; and (2) as a 2 × 2 factorial parameterization in a 2-piece spline model that tests for synergism between PRE and PRO treatments.

Two-Piece Spline Mixed-Effect Regression Models to Detect Differences Among Treatments in Milk Production

A mixed-effect regression model was used to evaluate the effect of treatment (independent variables) on the 4 milk outcomes (dependent variables Y_{ijk}) for 4 total models: the adjusted milk yield data using ECM milk, monthly milk fat yield, monthly milk protein yield (resulting table not shown), and nonadjusted daily milk yield. These are repeated measures because

Table 3. Marginal mean ECM (kg/d) estimated from the 4-level treatment mixed effects model (CON, PRE, PRO, SYN)

Conventional parameterization	Margin	SEM	Z-value[1]	P-value[1]	95% CI
CON	39.71	0.36	109.18	<0.01	39.00, 40.43
PRE	40.52	0.37	109.76	<0.01	39.78, 41.26
PRO	40.57	0.38	106.83	<0.01	39.85, 41.30
SYN	40.72	0.36	111.74	<0.01	40.00, 41.43

[1]Z-test and P-value for testing the null hypothesis that the marginal mean $\mu = 0$.

they are daily or monthly milk measures from the same cow over multiple lactations.

The model is

$$Y_{ijk} = \beta_0 + \Sigma_{a=1}^{3}\beta_a \text{Treatment}_k + \Sigma_{(a=4)}^{5}\beta_a \text{Parity}_{ijk} + \beta_6 \text{DIMprePK}_{ijk} + \beta_7 \text{DIMpostPK}_{ijk} + \beta_8 \text{DIMprePK}^2_{ijk} + \beta_9 \text{DIMpostPK}^2_{ijk} + \beta_{10}\text{Enrollment season}_k + \beta_{11}\text{Calving season}_k + \beta_{12}\text{Calf birth category}_{jk} + \beta_{13}\text{Pen group}_{ijk} + u_{0k}^{\text{Cow}} + u_{0k}^{\text{Parity}} + u_{1jk}^{\text{Parity}}\text{DIMprePK}_{ijk} + u_{2jk}^{\text{Parity}}\text{DIMpostPK}_{ijk} + e_{ijk}.$$

The model regresses the milk variable yield on the ith DIM of the jth parity of the kth cow (Y_{ijk}) on intercept β_0, the baseline of the overall milk variable yield at peak (PK). The β_1 to β_{13} regression coefficients represent the fixed effects in the models. The terms u_{0k}^{Cow} and u_{0jk}^{Parity} represent the random intercepts for cow and parity, respectively. The random slopes are represented by $u_{1jk}^{\text{Parity}}\text{DIMprePK}_{ijk}$ and $u_{2jk}^{\text{Parity}}\text{DIMpostPK}_{ijk}$ for DIM pre-peak and post-peak, respectively, which are the splines in the models. Residual error was represented as e_{ijk}.

Fixed Effects

The fixed effects for categorical variables are treatment, parity, season of enrollment, season of calving, calf birth category (including sex and twins and single births), and pen group (high and low milk production pens). The coefficients for these variables represent differences in the milk variables compared with the referent.

Treatment fixed effects ($\beta_1\text{Treatment}_1$, $\beta_2\text{Treatment}_2$, $\beta_3\text{Treatment}_3$) are PRE, PRO, and SYN, with CON employed as the referent group. Parity fixed effects are $\beta_4\text{Parity}_1$ and $\beta_5\text{Parity}_2$ for the second and third and greater parities of the kth cow, designating first parity as the referent.

Enrollment season fixed effect is β_{10}Enrollment_season$_k$ for the kth cow. In the original Lucey calf study, calf enrollment took 6 mo and included 3 periods: winter, spring, and summer. Following the seasonal calendar, winter and spring are combined as the referent level for this variable and compared with summer.

Season of calving fixed effect (β_{11}Calving_season$_k$ for the kth cow) is based on the seasonal calendar: winter (December 21 to March 20), spring (March 21 to June 20), summer (June 21 to September 22), and fall (September 23 to December 20). The summer and fall seasons are combined as the referent level and compared with the winter and spring seasons.

Calf birth category (β_{12} Calf_birth_catagory$_k$ for the kth cow) is assigned as either a single calf (non-twins, either sex), which includes both male and female, or same-sex twins, which includes male-male twins or female-female twins; or male-female twins as a separate dummy variable. The single calf was used as the referent category for comparison. We achieved a better model fit (lower AIC) by including the fixed effect calf birth category, a composite variable summarizing birth sex and twin status.

Pen group (β_{13}Pen_group$_{ijk}$ for the ith DIM, jth parity, and kth cow) represents cow movement between pens within a lactation. In early lactation, cows were in a high milk–producing pen, and then in later lactation they were moved to a lower milk–producing pen.

The fixed effects for continuous variables are days in milk pre-peak ($\beta_6\text{DIMprePK}_{ijk}$), which is the slope of the spline before reaching the peak of milk variable yield (spline 1), and the slope of the spline representing post-peak ($\beta_7\text{DIMpostPK}_{ijk}$) milk variable yield (spline 2). Both DIM pre-peak and DIM post-peak

Table 4. Effect differences for ECM (kg/d) estimated from the 4-level treatment mixed effects model (CON, PRE, PRO, SYN)

Conventional parameterization	Contrast	SEM	Z-value	P-value	95% CI
Prebiotic vs. control	0.81	0.52	1.57	0.12	−0.20, 1.8
Probiotic vs. control	0.86	0.51	1.69	0.09	−0.14, 1.9
Synbiotic vs. control	1.00	0.50	1.99	<0.05	0.015, 2.0

Table 5. Marginal mean ECM (kg/d) estimated from the 2 × 2 factorial mixed-effects model (prebiotic: yes/no; probiotic: yes/no)

2 × 2 parameterization						
Pre (0/1)	Pro (0/1)	Margin	SEM	*Z*-value[1]	*P*-value[1]	95% CI
0	0	39.71	0.36	109.18	<0.01	39.00, 40.43
1	0	40.57	0.37	109.76	<0.01	39.85, 41.30
0	1	40.52	0.38	106.83	<0.01	39.78, 41.26
1	1	40.72	0.36	111.74	<0.01	40.00, 41.43

[1]*Z*-test and *P*-value for testing the null hypothesis that the marginal mean $\mu = 0$.

meet at the peak of milk variable yield for each parity. The coefficients of these variables estimate the change in the milk yield given a 1-unit change in the continuous variable.

The last fixed effects for continuous variables are the 2 quadratic (squared) splines for pre- and post-peak slopes. The DIM pre-peak slope quadratic effect $\left(\beta_8 \mathrm{DIMprePK}^2_{\mathrm{ijk}}\right)$ represents the pre-peak milk variable from the time of calving to the peak of milk yield variable. The DIM post-peak slope quadratic $\left(\beta_9 \mathrm{DIMprePK}^2_{\mathrm{ijk}}\right)$ represents the post-peak milk yield variable from peak milk to dry-off. Using 2 splines in the models allows the peak of milk yield variable to differ for each cow parity, and the squared term allows each spline to be a curve representing a complete lactation curve.

Random Effects

The random effects capture the variability in milk yield in the ith DIM $\left(\mathrm{u}^{\mathrm{Parity}}_{1\mathrm{jk}} \mathrm{DIMprePK}_{\mathrm{ijk}} \text{ and } \mathrm{u}^{\mathrm{Parity}}_{1\mathrm{jk}} \mathrm{DIMprePK}_{\mathrm{ijk}}\right)$ of the jth parity $\left(\mathrm{u}^{\mathrm{Parity}}_{0\mathrm{jk}}\right)$ of the kth cow $\left(\mathrm{u}^{\mathrm{Cow}}_{0\mathrm{k}}\right)$ with the residual error ($\mathrm{e}_{\mathrm{ijk}}$).

The random effects capture the variability in milk yield in the ith DIM $\left(\mathrm{u}^{\mathrm{Parity}}_{1\mathrm{jk}} \mathrm{DIMprePK}_{\mathrm{ijk}} \text{ and } \mathrm{u}^{\mathrm{Parity}}_{1\mathrm{jk}} \mathrm{DIMprePK}_{\mathrm{ijk}}\right)$ of the jth parity $\left(\mathrm{u}^{\mathrm{Parity}}_{0\mathrm{jk}}\right)$ of the kth cow $\left(\mathrm{u}^{\mathrm{Cow}}_{0\mathrm{k}}\right)$ with the residual error ($\mathrm{e}_{\mathrm{ijk}}$).

These random spline effects capture the variability in milk production (random slope) pre- and post-peak for each parity random intercept, nested within random cow intercept, adjusting for repeated measures within lactation and across parities of a cow. Two random slope coefficients were used to shape milk lactation curve in each cow's parity using DIM pre-peak and DIM post-peak. The peak was considered the highest milk yield within a parity. The DIM pre-peak was calculated as follows:

$$(\mathrm{DIM} - \mathrm{DIM\ at\ peak})/\mathrm{DIM\ at\ peak},$$

whereas the DIM post-peak was calculated as

$$(\mathrm{DIM} - \mathrm{DIM\ at\ peak})/(\mathrm{total\ DIM} - \mathrm{DIM\ at\ peak}).$$

All random effects were assumed to be independent and identically distributed with a mean of 0 and variances cow and parity, with their values estimated using empirical best linear unbiased predictor. At the parity level, pre-peak and post-peak milk, milk fat, and milk protein yields were grouped together and assumed to follow a multivariate normal distribution. This distribution had a mean of 0 and a 3 × 3 unstructured variance-covariance matrix.

The final models were built using manual forward selection. A change of 20% or more in the ECM yield in the estimate of treatment effect with and without potential confounders (parity, enrollment season, calf birth category, or calving season) in the final model was deemed as evidence of confounding. None of the fixed effects parity, pen group, or calf sex had a confounding effect at the 20% threshold. However, the PRO treatment was confounded by the season of enrollment and calving season. Thus, those variables were included in the model to control for their confounding effects. Two-way interactions between independent variables were examined. No two-way interactions were identified between the treat-

Table 6. Effect differences for ECM (kg/d) estimated from the 2 × 2 factorial mixed-effects model (prebiotic: yes/no; probiotic: yes/no)

2 × 2 parameterization	Contrast	SEM	*Z*-value	*P*-value	95% CI
Prebiotic vs. control	0.81	0.52	1.57	0.12	−0.20, 1.8
Probiotic vs. control	0.86	0.51	1.69	0.09	−0.14, 1.9
Synbiotic vs. control	1.00	0.50	1.99	<0.05	0.015, 2.0

Table 7. Two-piece spline mixed-effect regression model to detect differences among treatments in ECM milk yield

Factor	Coefficient (ECM kg/d)	SEM	*P*-value	95% CI
Treatment				
Control	Referent			
Prebiotic	0.81	0.5	0.1	−0.20, 1.8
Probiotic	0.86	0.5	0.09	−0.14, 1.9
Synbiotic	1.0	0.5	<0.05	0.015, 2.0
Parity				
1	Referent			
≥2	12.4	0.3	<0.05	12, 13
Season of enrollment				
Fall–winter	Referent			
Summer	−2.0	0.7	<0.05	−3.3, −0.70
Calving season				
Summer–fall	Referent			
Winter–spring	0.97	0.34	<0.05	0.31, 1.6
Pen group				
Low	−1.5	0.1	<0.05	−1.7, −1.3
Calf birth category				
Single (male or female)	Referent			
Twins (same sex)	0.14	0.8	0.9	−1.4, 1.7
Twins (different sex)	−1.1	0.9	0.3	−2.9, 0.76
DIM pre-peak	11	0.6	<0.05	9.4, 12
DIM post-peak	−20	0.5	<0.05	−21, −19
DIM pre-peak squared	−2.8	0.7	<0.05	−4.2, −1.5
DIM post-peak squared	−0.46	0.5	0.3	−1.4, 0.48
Peak milk (kg; intercept)	42	0.5	<0.05	41, 43
Random effect[1]				
Cow intercept	23	2		19, 27
Parity intercept	27	1		25, 30
DIM pre-peak slope	23	2		19, 28
DIM post-peak slope	140	5		130, 150
Overall error	30	0.3		29, 31

[1]Random effect estimates represent variance components.

ments and fixed effects (parity, enrollment season, pen group, and calving season), and no interaction effect was found between treatment and calf sex (male, female) or treatment and twin status on ECM yield. Best-fit models were chosen using the lowest AIC and LRT <0.5. The locally weighted scatterplot smoothing (LOWESS) function (Stata BE18.0) was used to plot the effects of CON, PRE, PRO, and SYN on the milk variable, adjusted to 4% fat and 3.3% protein, over 305 DIM in parities 1 to 3.

A similar 2-piece splines mixed-effects model was used to analyze the same dataset using the 2 × 2 factorial parametrization that represented treatment as prebiotic (yes/no) or probiotic (yes/no) and evaluated whether the SYN response can be explained by additive main effects and a prebiotic × probiotic interaction as a departure from additivity. Splines, random effects structure, model building, and assessing model fit followed the same process described previously.

RESULTS AND DISCUSSION

The number of cows in the study was distributed equally across 4 treatments: 333 in CON, 314 in PRE, 319 in PRO, and 330 in SYN at the beginning of their first lactation. The numbers of lactations (26%, 24%, 25%, and 26% of total), the numbers of ECM milk records (25%, 24%, 25%, and 26% of total), and the numbers of nonadjusted milk records (25%, 24%, 25%, 26% of total) per treatment were similar for CON, PRE, PRO, and SYN, respectively. Because this study was performed with a commercial dairy herd, there was no way to balance treatments for season of enrollment.

Heifer Survival Outcomes

Figure 1 shows the numbers of calves and cows that began the study and were included in each follow-up analyses. Out of the 1,801 calves in the original study (Lucey et al., 2021), 1,296 made it to their first lactation and 527 made it to their third lactation.

In the group of study females that calved for the first time (n = 1,296), culling and death rates also did not differ between treatments ($P > 0.05$; Table 2). However, parity exhibited a significant association with culling and death ($P < 0.01$). Cows in their third lactation had lower odds of exiting the herd compared with first-lactation cows ($P < 0.05$), which is expected because the number of cows in the herd decreased as lactation number in-

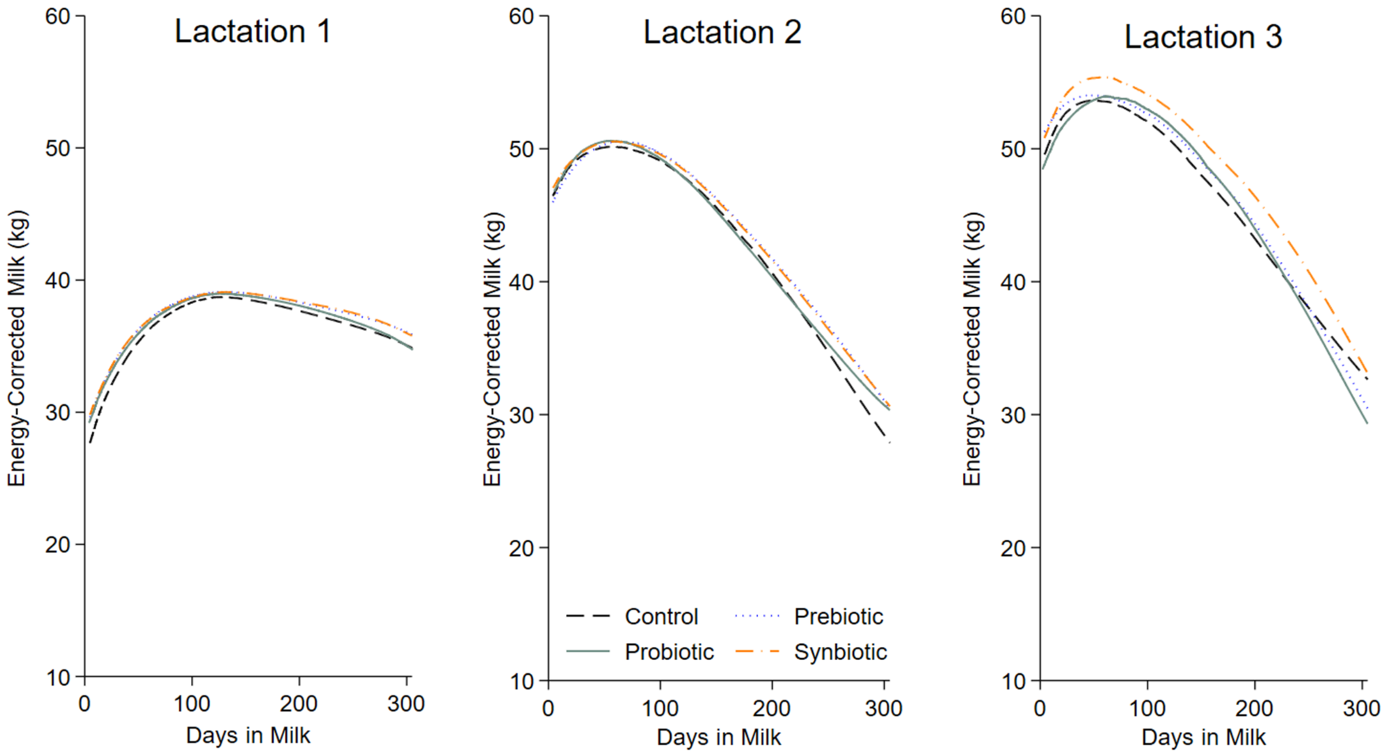

Figure 2. Locally weighted scatterplot smoothing (LOWESS) illustrating the effects of control, prebiotic, probiotic, and synbiotic treatments on ECM adjusted to 4% fat and 3.3% protein. Graphs represent ECM trends over 305 DIM for parities 1 through 3.

creased (Figure 1). Parity-4 cows had an average of only 3.6 DHIA tests, as cows had not completed their lactation at the time of data collection. Therefore, parity-4 records were excluded from the analysis, resulting in a total of 2,735 lactations for the final analysis.

Contrasting Treatment Effects Under the Conventional Versus 2 × 2 Factorial Parametrizations

The factorial analysis indicated that both prebiotic and probiotic supplementation were associated with modest increases in ECM, but the synbiotic response did not exceed the additive expectation of these main effects. Consequently, no statistical evidence of synergy between prebiotic and probiotic components was detected, despite the numerically higher ECM observed for synbiotic-treated cows.

The relationships between the 2 modeling approaches are presented as marginal mean estimates from both the conventional treatment (CON, PRE, PRO, and SYN) parameterization (Tables 3 and 4) and the 2 × 2 factorial parameterization (Tables 5 and 6). Both models were fitted to the same data and include identical covariates and random-effects structures. The adjusted means were identical across the 2 parameterizations, as expected, given that both models are fitted to the same dataset with the same covariates, spline terms for DIM, and random-effects structure; the only difference was how the treatment effects were parameterized (PRE, PRO, and SYN treatment indicators vs. prebiotic main effect, probiotic main effect, and their interaction). Both models had identical estimates and showed that calves supplemented with SYN had higher milk production as adult cows compared with their control counterparts, yet not higher than expected under the 2 × 2 factorial additivity, and therefore the SYN effect was not synergistic. Both models estimated the difference between synbiotic and control at +1.0 kg ECM/d, indicating that the magnitude of the SYN effect is consistent across the 2 parameterizations. The study's mixed-effect regression model using 2-piece splines was novel in its lactation-specific peak that varied across parities and by cow, compared with a fixed-DIM peak. Both model parametrizations served different purposes. The conventional parametrization of treatment effects was specified to directly estimate the treatment effects on milk production compared with the controls yet does not test specifically for synergism; the latter was achieved using a model with treatment effects specified under a 2 × 2 factorial parametrization (see Appendix Table A1 for 2 × 2 table).

Table 8. Statistical model for monthly milk fat yield, including results for fixed and random effects, using a 2-piece mixed-effects spline regression

Factors	Coefficient (milk fat kg/d)	SEM	*P*-value	95% CI
Treatment				
Control	Referent			
Prebiotic	0.034	0.02	0.2	−0.012 0.081
Probiotic	0.033	0.02	0.2	−0.013, 0.079
Synbiotic	0.048	0.02	<0.05	0.0024, 0.094
Parity				
1	Referent			
≥2	0.55	0.01	<0.05	0.52, 0.57
Season of enrollment				
Fall–winter	Referent			
Summer	−0.11	0.03	<0.05	−0.17, −0.050
Calving season				
Summer–fall	Referent			
Winter–spring	0.053	0.02	<0.05	0.021, 0.085
Pen group				
Low	−0.054	0.007	<0.05	−0.068, −0.040
DIM pre-peak	1.1	0.04	<0.05	0.99, 1.1
DIM post-peak	−1.5	0.03	<0.05	−1.5, −1.4
DIM pre-peak squared	0.53	0.04	<0.05	0.45, 0.61
DIM post-peak squared	0.50	0.03	0.3	0.45, 0.56
Peak fat (kg; intercept)	1.9	0.2	<0.05	1.8, 1.9
Random effect[1]				
Cow intercept	0.044	0.004		0.037, 0.052
Parity intercept	0.061	0.003		0.054, 0.067
DIM pre-peak slope	0.046	0.007		0.035, 0.062
DIM post-peak slope	0.26	0.01		0.24, 0.28
Overall error	0.11	0.001		0.11, 0.12

[1]Random effect estimates represent variance components.

Effects of Treatments on ECM Yield

The mixed-effect regression model examining treatment effects on ECM yield (Table 7) showed that SYN cows had increased ECM yield compared with CON. Multiparous cows produced more ECM. Cows born in the summer (enrollment) season produced less ECM as adults compared with cows born in the fall and winter seasons. Cows that calved in winter and spring also had higher ECM production. Other studies conducted in similar climates to California have found similar results. Chester-Jones et al. (2017) from the University of Minnesota found that calves born in summer produced ~0.7 kg/d more 305-d adjusted milk in their first lactation than calves born in other seasons. Ray et al. (1992) examined cow records from Arizona and found that cows which calved in winter and spring had 1 kg/d lower 305-d fat-corrected milk production compared with cows that calved in summer. In a study from Norman et al. (1995) conducted in California, milk yield was lowest for cows calving in summer and highest for fall and winter.

These results are also consistent with previous study findings by Chester-Jones et al. (2017) from the University of Minnesota; they also found that calves with higher ADG at 6 wk produced more 305-d adjusted milk in their first lactation. The positive relationship between early-life health, growth, and future milk production observed in this study could be explained by the effect of SYN supplementation on gastrointestinal function during the preweaning period. Synbiotics enhance gut colonization due to their ability to introduce beneficial bacteria and then support growth with a prebiotic. This may help colonization of the gastrointestinal tract, which may have a lasting effect of increasing milk production (Rioux et al., 2005). Soberon et al. (2012) found that, for every 1-kg increase in ADG during the preweaning period, milk yield increased by ~850 kg in the first parity in a cohort of 1,800 preweaning calves. Similarly, Jiang et al. (2025), who conducted a meta-analysis of 18 studies, found that every 100 g/d increase in ADG during the preweaning period was associated with an increase of 302 kg in milk yield, 10.6 kg in milk fat, and 11.1 kg in milk protein during the first lactation. These results align with our findings, where ECM increased in response to SYN despite the absence of a synergistic interaction between prebiotic and probiotic components.

Modeling ECM using splines transformed the intercept to represent the peak of ECM during the cows' parities. Pre-peak and post-peak splines, along with their squared terms, represented the relationship between ECM yield and time (DIM) on a transformed scale and captured the impact of these 2 phases on ECM for different parity periods. The splines captured the progression of milk ECM yield during the parity, allowing the model to describe

Table 9. Statistical model for nonadjusted (daily) milk yield, including results for fixed and random effects, using a 2-piece mixed-effects spline regression

Factors	Coefficient (milk kg/d)	SEM	*P*-value	95% CI
Treatment				
Control	Referent			
Prebiotic	0.66	1	0.3	−0.61, 1.9
Probiotic	1.1	2	0.08	−0.13, 2.4
Synbiotic	1.3	2	<0.05	0.0090, 2.5
Parity				
1	Referent			
≥2	13	0.3	<0.05	13, 14
Season of enrollment				
Fall–winter	Referent			
Summer	2.5	2	0.1	−0.64, 5.7
Calving season				
Summer–fall	Referent			
Winter–spring	1.4	0.4	<0.05	0.64, 2.2
Interaction of treatment and enrolling season				
Prebiotic and summer	−1.7	2	0.5	−6.3, 2.9
Probiotic and summer	−5.3	2	<0.05	−9.7, −0.92
Synbiotic and summer	−4.7	2	<0.05	−9.2, −0.19
DIM pre-peak	−15	0.2	<0.05	−15, −14
DIM post-peak	−3.9	0.3	<0.05	−4.5, −3.4
DIM pre-peak squared	−31	0.2	<0.05	−32, −31
DIM post-peak squared	−16	0.1	<0.05	−16, −16
Peak milk (kg; intercept)	35	0.6	<0.05	34, 36
Random effect[1]				
Cow intercept	35	3		30, 41
Parity intercept	50	2		46, 54
DIM pre-peak slope	100	3		95, 110
DIM post-peak slope	170	5		160, 180
Overall error	40	0.1		40, 40

[1]Random effect estimates represent variance components.

the ECM dynamics across DIM. For more information on the mixed-effect spline model, a detailed example is presented in the Appendix, and Figure 2 shows the ECM yield calculation across the time spline.

Effects of Treatments on Milk Fat and Milk Protein Yields

To examine the increase in ECM from cows that were fed SYN as preweaning calves, the same 2-piece mixed-effects spline model was used to examine milk protein yield and milk fat yield separately. Milk protein yield was not different between treatments (results not shown), but milk fat yield was increased in SYN compared with CON and accounted for most of the increase in ECM (Table 8). Multiparous cows produced more milk fat. Cows born in the summer season produced less milk fat compared with cows born in the fall–winter season. Also, cows that calved in the winter–spring season produced more milk fat compared with cows that calved in the summer–fall season. Differences in the DIM pre-peak, post-peak, pre-peak squared, and post-peak squared reflect the different pattern of milk fat yield throughout the lactation compared with ECM yield.

Effects of Treatments on Nonadjusted Milk Yield

The 2-piece mixed-effect spline regression model was also employed to evaluate the effects of different treatments on nonadjusted milk yield (Table 9). Similar results to the adjusted ECM yield model were found, where SYN had an increased nonadjusted milk yield compared with CON; however, PRO had a tendency for increased nonadjusted milk yield. Multiparous cows produced more milk compared with first-lactation cows, and pre- and post-peak DIM had lower nonadjusted milk yield. Cows that calved during the winter–spring season had increased nonadjusted milk yield compared with cows that calved in the summer–fall. The PRO and SYN cows had decreased nonadjusted milk yield in the summer season compared with CON.

CONCLUSIONS

This study showed that feeding a combination of *Saccharomyces cerevisiae* yeast culture, *Bacillus subtilis*, and *Lactobacillus plantarum* during the preweaning period increased ECM yield (or fat and milk yields) in adult lactating Holstein cows. The increase in ECM yield could be explained by significantly higher fat content in milk produced by cows fed the combination supplement during their preweaning period, compared with control cows.

NOTES

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Nonstandard abbreviations used: AIC = Akaike information criterion; CON = control group; IQR = interquartile range; LRT = likelihood ratio test; PRE = prebiotic, 7 mL of *Saccharomyces cerevisiae* yeast culture; PRO = probiotic, *Bacillus subtilis* and *Lactobacillus plantarum*, delivering ~1 billion and 250 million cfu per head per day, respectively; Q3 = third quartile; SYN = synbiotic, a combination of both PRE and PRO at the same dosages as the PRE and PRO treatments.

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APPENDIX

Example of Spline Analyses of the Parity Curve

In this example, we illustrate how the ECM yield is calculated for a 305-d parity starting at DIM 1 (−1 on spline-transformed timeline), peak (0 on spline-transformed timeline), and DIM 305 (1 on spline-transformed timeline) in the first and subsequent parities (parities ≥2). The plot of production against the spline-transformed timeline is projected to reflect equidistance from −1 to 0 and 0 to +1. The parity timeline with peak milk occurred

Table A1. Table of 2 × 2 factorial analyses with prebiotic (yes/no) and probiotic (yes/no) main effects and their interaction, to detect differences among treatments in ECM milk yield

Factor	Coefficient (ECM kg/d)	SEM	*P*-value	95% CI
Treatment				
Prebiotic	0.81	0.5	0.1	−0.20, 1.8
Probiotic	0.86	0.5	0.09	−0.14, 1.9
Prebiotic and probiotic	−0.67	0.7	0.4	−2.1, 0.75
Parity				
≥2	12.4	0.3	<0.05	12, 13
Season of enrollment				
Summer	−2.0	0.7	<0.05	−3.3, −0.70
Calving season				
Winter–spring	0.97	0.34	<0.05	0.31, 1.6
Pen group				
Low	−1.5	0.1	<0.05	−1.7, −1.3
Calf birth category				
Twins (same sex)	0.14	0.8	0.9	−1.4, 1.7
Twins (different sex)	−1.1	0.9	0.3	−2.9, 0.76
DIM pre-peak	11	0.6	<0.05	9.4, 12
DIM post-peak	−20	0.5	<0.05	−21, −19
DIM pre-peak squared	−2.8	0.7	<0.05	−4.2, −1.5
DIM post-peak squared	−0.46	0.5	0.3	−1.4, 0.48
Peak milk, kg (intercept)	42	0.5	<0.05	41, 43
Random effect[1]				
Cow intercept	23	2		19, 27
Parity intercept	27	1		25, 30
DIM pre-peak slope	23	2		19, 28
DIM post-peak slope	140	5		130, 150
Overall error	30	0.3		29, 31

[1]Random effect estimates represent variance component.

earlier, before the midpoint of parity, and produced the classical parity curve shape. The following equation can be used to estimate the ECM yield:

$$Y_{ijk} = \beta_0 + \beta_1 \text{DIMprePK}_{ijk} + \beta_2 \text{DIMpostPK}_{ijk} + \beta_3 \text{DIMprePK}^2_{ijk} + \beta_4 \text{DIMpostPK}^2_{ijk},$$

where β_0 is the intercept, representing the peak milk yield; $\beta_1\text{DIMprePK}_{ijk}$ + $\beta_2\text{DIMpostPK}_{ijk}$ are the coefficients for the linear spline terms; and $\beta_3\text{DIMprePK}^2_{ijk} + \beta_4\text{DIMpostPK}^2_{ijk}$ are the coefficients for the squared terms, capturing curvature.

For parity 1, the ECM can be calculated as follows for the 3 key points DIM 1 (pre-peak), peak (DIM 0 on the time spline axis), and DIM 305 (post-peak).

For DIM 1 (pre-peak),

$$\text{ECM} = 42.1 + [10.6 \times (-1)] + [-2.84 \times (-1)^2] = 28.66 \text{ kg}.$$

This result indicates that the estimated ECM for DIM 1 (−1 on the spline-transformed timeline) for cows in their first parity corresponds to 28.66 kg on the DIM scale.

Peak (DIM 0): The estimated peak ECM for the cows in the (DIM 0) on the first parity was 42.1 kg.

For DIM 305 (post-peak),

$$\text{ECM} = 42.1 + [(-20.30) \times (+1)] + [(-0.46) \times (+1)^2] = 21.34 \text{ kg}.$$

The estimated ECM yield by the end of the parity is 21.34 kg.

For parity ≥2, an additional term ($\beta_5\text{Parity}_{jk}$) is added to the formula to account for the increase in production for later parities:

$$Y_{ijk} = \beta_0 + \beta_1 \text{DIMprePK}_{ijk} + \beta_2 \text{DIMpostPK}_{ijk} + \beta_3 \text{DIMprePK}^2_{ijk} + \beta_4 \text{DIMpostPK}^2_{ijk} + \beta_5 \text{Parity}_{jk}.$$

For DIM 1 (pre-peak),

$$\text{ECM} = 42.1 + [10.6 \times (-1)] + [(-2.84) \times (-1)^2] + 12.4 = 41.06 \text{ kg}.$$

Peak (DIM 0),

$$\text{ECM} = 42.1 + 12.4 = 54.2 \text{ kg}.$$

For DIM 305 (post-peak),

$$\mathrm{ECM} = 42.1 + [(-20.30) \times (+1)] + [(-0.46) \times (+1)^2] + 12.4 = 33.74 \text{ kg.}$$

For milk yield estimates at other DIM, the respective fraction on the spline-transformed timeline is used in the model prediction to estimate milk yield at a given DIM. These calculations illustrate how the spline-transformed timeline and coefficients predict ECM yield at different DIM during parity, offering a detailed representation of the ECM estimates on the parity curve.